

LINE-UP & TAKE-OFF

Normal take-off procedure - Line-up calls - Departure briefing - Climb-out sequence

P-Mentor / Tecnam P2002 Stage: DGH 03 VFR - Day EKRK Circuit Briefing: 10 min

Lesson Timing					
10 MIN Briefing		1.0 HR Flight		15 MIN Debrief	
PREREQUISITES					
- Ground school: Pre-flight inspection, taxiing (DGH 01-02) - Familiarity with aircraft cockpit and controls (P-Mentor) - Completed pre-flight brief and performance calculations - Student has studied SOP sections 1.9 and 1.11					
LESSON OBJECTIVES					
- Deliver and execute a complete departure briefing per SOP 1.9.1 - Perform correct line-up procedure and calls before RWY entry - Execute normal take-off from brake release to 70 KIAS climb - Apply correct flap retraction sequence at 400 ft AGL - Establish cruise climb and complete after take-off checklist					
TOLERANCES					
53 kt	70 kt	400 ft	+5 m	+10 deg	500 ft
VR CALL	VY CLIMB	FLAPS AGL	CENTRELINE	CLIMB HDG	TURN AGL
MOTIVATION					
The take-off is the first student-executed manoeuvre of the flight training programme. It establishes the foundation for all departure and circuit work. Correct procedure, positive control inputs, and verbal call-outs from the outset build habits that persist throughout the student's flying career.					
COMMON MISTAKES					
- Departure briefing incomplete or engine failure plan not operationally stated - Missing "Final clear" or HSI check during line-up procedure - Insufficient right rudder causing directional deviation on roll - No VR call-out at 53 KIAS, or late/slow rotation - Flap retraction without speed call or below safe speed - MCP not set to 2250 RPM after flap retraction					
Source Material					
SOP	Normal Take-Off Procedure - P-Mentor				§ 1.11.1
SOP	Departure Briefing Standard				§ 1.9.1
SOP	Line-Up Procedure				§ 1.10
SOP	Engine Failure After Take-Off				§ 1.12

ATP

Take-Off & Climb Lesson Prep

ATP INT

01 ENVIRONMENT

Operating area - VMC minima - Communications - Circuit area

THEORY

Operating Environment - EKRK

- Circuit area: EKRK Roskilde Airport. Uncontrolled aerodrome (AFIS).
- VMC minima Day VFR: 1000 ft cloudbase above circuit height, 5 km visibility.
- Wind preference: calm to light wind for first exposure. XWC max 5 kt.
- COM: EKRK AFIS 120.225 MHz. Check ATIS before departure.
- QNH: Set and cross-check on both altimeters before taxi.
- Active runway determined by AFIS - confirm before departure brief.

SOP

§ 0.5

Aerodrome operating procedures, frequencies and noise abatement

02 SETUP

Departure briefing - Line-up procedure - HSI check

PROCEDURE

Departure Briefing - SOP 1.9.1

KEY PRINCIPLE

Brief BEFORE entering the hold. Every take-off requires a full departure briefing. Include runway, SID/visual departure, engine failure plan, and hand-over.

STANDARD DEPARTURE BRIEFING - (SOP 1.9.1)

- 1 State runway: "We are departing RWY [XX]. Wind [XXX]/[XX] kt, QNH [XXXX]."
- 2 State departure: "Departure [visual/SID]. Climbing to [altitude]. Turning [left/right/straight ahead]."
- 3 State engine failure plan: "In the event of engine failure before VR - throttle idle, brakes as required, runway straight ahead. After VR below 500 ft AGL - nose down 70 kt, land within 45 deg. Above 500 ft - [weather/circuit permitting] attempt return, otherwise land ahead."
- 4 State hand-over: "I have control for take-off. I will hand over at [altitude/point]."

SOP

§ 1.9.1

Departure briefing is mandatory before EVERY take-off, including touch-and-go circuits

PROCEDURE

Line-Up Procedure - SOP 1.10

HOLDING POINT CHECKS (§ 1.9.1)

- 1 Complete BEFORE TAKE-OFF checklist in the hold.
- 2 Set DI (HSI) to compass. Set RWY heading bug.
- 3 Report ready: "EKRK, [callsign], Holding [point], Ready RWY [XX]."
- 4 After ATC/AFIS clearance: check final approach for traffic.
- 5 Call out: "Final clear - Lining up RWY [XX]"

LINE-UP CHECKS (§ 1.10)

- 6 Enter runway. On uncontrolled: scan base and final before entering.
- 7 Align with centreline. Apply full right rudder, release, centralise.

8

Check HSI - verify on correct RWY heading. Call out: "Heading indicator checked - RWY [XX] identified"

9

Select reference point on extended centreline for track reference.

SOP

§ 1.10

Wake turbulence: Twin Otter (medium) - start timer at rotation of preceding traffic. Wait min 90 sec before reporting ready

03 EXECUTION

Take-off roll - Rotation - Initial climb - Flap retraction - MCP

THEORY

Take-Off Aerodynamics & Control

- Engine torque and P-factor yaw the nose left on full power application - right rudder required throughout the roll.
- Slight forward pressure on control wheel during roll keeps nose wheel on surface, improving directional control.
- At VR (53 KIAS) wings generate sufficient lift to fly - rotation must be prompt and positive to 70 KIAS attitude.
- Flap 1 provides extra lift reducing ground roll distance and lowering VR.
- VX 56 KIAS = best angle of climb (obstacle clearance). VY 70 KIAS = best rate (normal climb).
- MCP 2250 RPM set after flap retraction - engine protection, noise abatement, efficient climb power.

MANOEUVRE

Normal Take-Off - P-Mentor (SOP 1.11.1)

KEY PRINCIPLE

Power on - directional control with rudder - let the aircraft fly itself off at 53 KIAS. Rotate promptly and positively. Think attitude and airspeed, never control input alone.

PHASE 1 - TAKE-OFF ROLL

- 1
- ATC cleared for take-off: switch LANDING light ON. Maintain TAXI light ON.
- 2
- Hold brakes. Advance throttle smoothly to MAP 20". Call out: "Engine instruments checked - Timing started"
- 3
- Release brakes. Advance throttle smoothly to full power.
- 4
- Call out: "Full power" - verify RPM and MAP at max.
- 5
- Apply right rudder to maintain centreline. Slight forward pressure on control wheel. Track reference point.

PHASE 2 - ROTATION & INITIAL CLIMB

- 6
- At 53 KIAS: Call out "53 - Rotate". Release forward pressure smoothly. Let aircraft fly off at 70 KIAS climb attitude.
- 7
- Establish and hold VY 70 KIAS climb attitude. Trim to relieve control pressure.
- 8
- Track extended centreline using reference point selected at line-up. Climb out straight.

PHASE 3 - FLAP RETRACTION & CLIMB-OUT

- 9
- At 400 ft AGL: check positive rate of climb, verify safe speed (min VX 56 KIAS). Call out: "Safe speed - Flaps up"
- 10
- Select flaps UP. Adjust pitch slightly. Retard propeller to 2250 RPM (MCP). Call out: "Max continuous power". Re-trim.
- 11
- At 500 ft AGL: lookout. Commence turn as briefed in departure briefing.
- 12
- Passing 1000 ft AGL: trim for 80 KIAS cruise climb. Select TAXI light OFF. Complete AFTER TAKE-OFF checklist from memory.

SOP		§ 1.11.1
SOP	No checklist reading below 1000 ft AGL except memory items (flap retraction)	§ 1.11.1
	Turbulence/gusts: delay VR by 5 kt or half gust factor. Climb at 70 kt flaps 1, then VY+5 after retraction	

WARNING

Engine Failure After VR

Below 650 ft MSL (500 ft AGL): nose down to 70 KIAS, throttle idle, land straight ahead within 45 deg of into-wind direction. Do NOT attempt to return to the field. Above 650 ft: standard emergency procedures. This scenario MUST be briefed BEFORE every take-off.

CAUTION**Flap Retraction**

ALWAYS verify safe speed (minimum VX 56 KIAS) and call out before selecting flaps UP. Never retract flaps if airspeed is insufficient - aircraft will sink. If positive rate of climb is not confirmed, do not retract flaps.

COMMON ERRORS - TAKE-OFF EXECUTION

- Insufficient right rudder - nose drifts left on power application
- Late rotation - aircraft skips or floats on runway
- Not maintaining 70 KIAS climb attitude after rotation
- Missing speed call before flaps up at 400 ft AGL
- Not tracking centreline reference point after lift-off
- Forgetting MCP 2250 RPM after flap retraction

FI TEACHING POINTS

Demo the full sequence on first attempt - talk through all steps aloud. Student takes over on 2nd take-off. Watch for feet on brakes during roll - common early error. Ensure student calls out each step; silence means not thinking. If student drifts off centreline, ask them to identify their reference point. Praise positive procedure language: "Rotate" not "pull back".

04 CHECK

Assessment criteria - Debrief points - Progression

ASSESSMENT

Lesson Assessment Criteria

SATISFACTORY - PROGRESS

- Departure briefing complete and operationally correct
- Line-up calls made correctly at each stage
- Centreline maintained during take-off roll (+5 m)
- VR call-out at 53 KIAS with prompt rotation
- 70 KIAS climb attitude established and maintained
- Speed call made before flap retraction at 400 ft AGL
- MCP set and centreline tracked on climb-out

UNSATISFACTORY - REPEAT

- Departure briefing missing or engine failure plan incomplete
- No "Final clear" call before RWY entry
- No HSI check on line-up
- Consistent directional control failures on take-off roll
- Flap retraction without speed call or at unsafe speed
- No VR call-out or consistently late rotation

53 kt

VR CALL

70 kt

VY CLIMB

400 ft

FLAPS AGL

+5 m

CENTRELINE

+10 deg

CLIMB HDG

500 ft

TURN AGL

DEBRIEF

Structured Debrief - 7 Points

- 1 Student self-assessment: "How did the take-off roll feel? Were you in control of direction at all times?"
- 2 Review departure briefing quality - was the engine failure plan operationally complete and correctly stated?
- 3 Line-up procedure - were all calls made? HSI checked? "Final clear" called before RWY entry?
- 4 Take-off roll - directional control quality, rudder use, reference point tracking. Any deviations?
- 5 Rotation and climb - was VR called at 53 KIAS? Did aircraft climb at 70 KIAS? Was trim applied?
- 6 Flap retraction sequence - speed call made? MCP set? Centreline tracked to 500 ft AGL turn?
- 7 Progress decision and FlightLogger entry. Review SOP 1.11.1 and 1.9.1 before next lesson.

PROGRESSION

Next Lesson - DGH 04 Circuit & Landing

- DGH 04: Normal circuit pattern and normal landing
- Student study: SOP 1.18.1 Normal Landing, circuit procedure, PAPI, final speeds
- Review: Go-around procedure SOP 1.20

Notice: This briefing follows the CAPA standard layout. SOP E2R1 is the authoritative reference. This package supplements, does not replace, the SOP. Printed copies are not controlled.